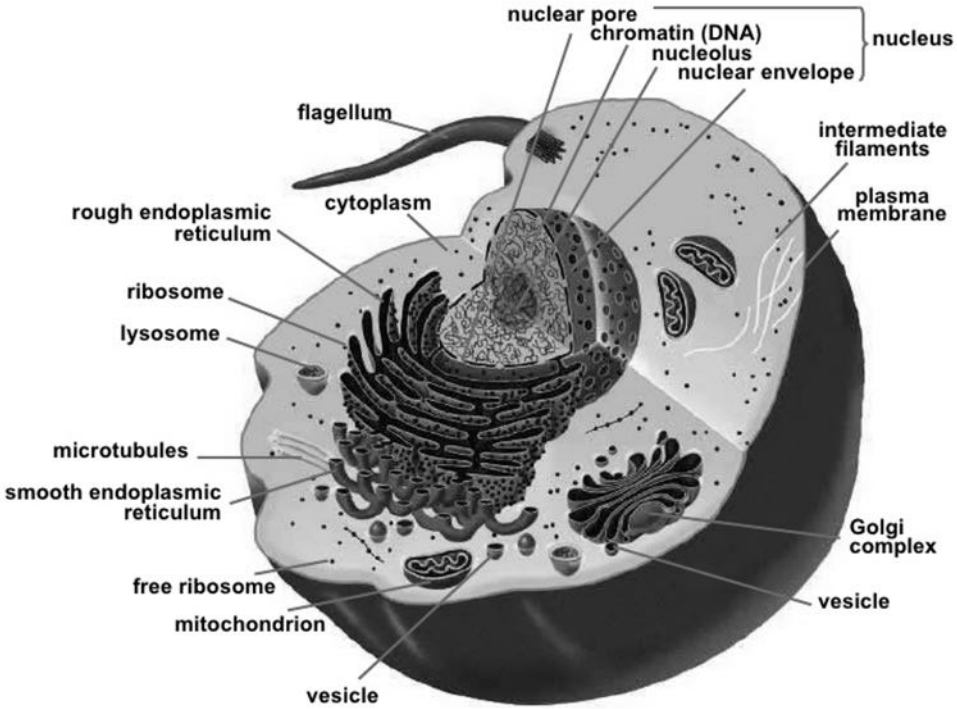
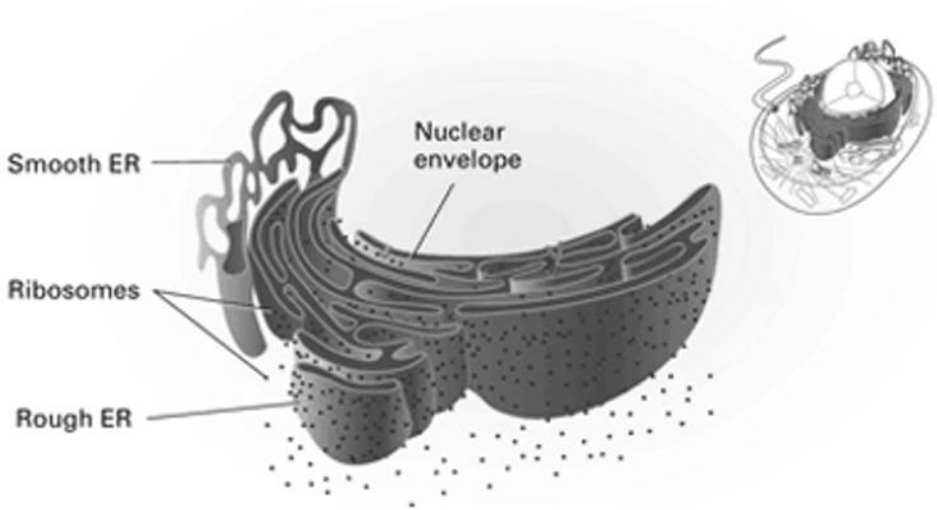
II. Endoplasmic Reticulum

It is a single membrane system of intercommunicating membranous channels in the form of tubules that serve a function involving protein synthesis and transport.

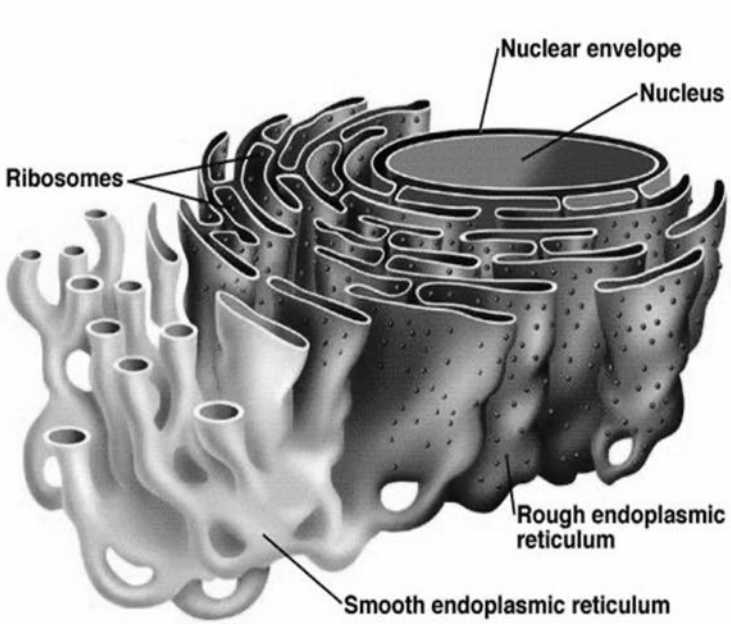




Endoplasmic Reticulum

Rough ER	Smooth ER

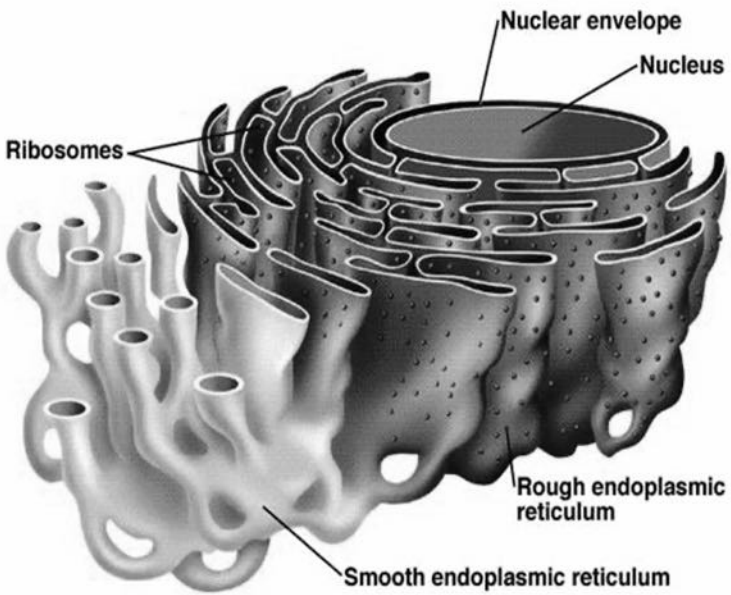
Rough ER (r-ER)	* Attached to ribosomes on its surface.
	* Synthesis, Segregation and delivery of protein.
	* Protects cytoplasm from enzymes.
	* Position: Fibroblast, Liver and Pancreatic cells.



Endoplasmic Reticulum

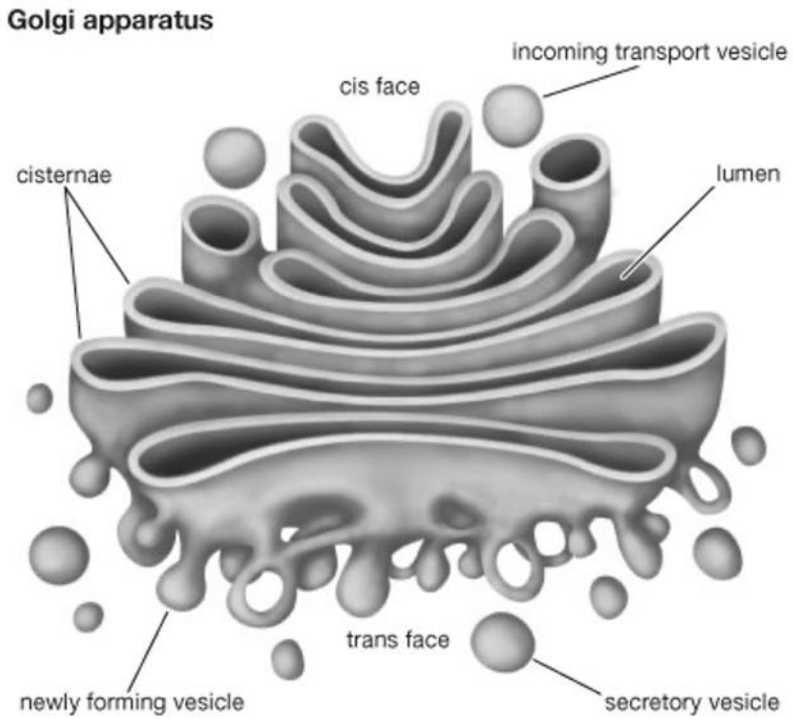
Rough ER	Smooth ER

Smooth ER (s-ER)	* Without ribosomes on its surface.
	* Synthesis of lipids and glycogen.
	* Plays a role in muscular contraction.
	* Position: Liver and Endocrine cells.



III. Golgi apparatus

Composed of 3 elements: Flattened stacks, Transfer vesicles and secretory vesicles.



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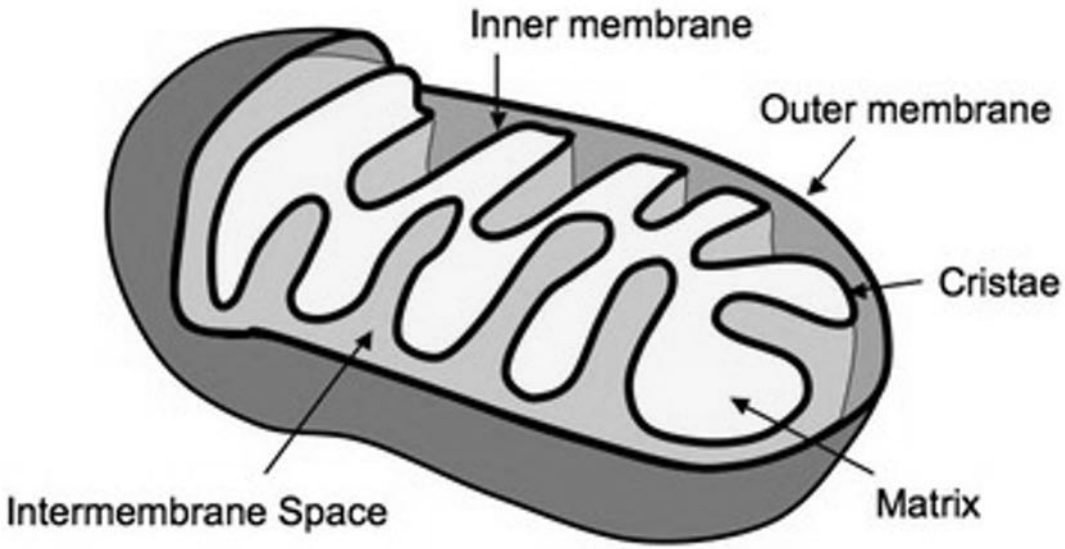
Concerned with the secretory activity of the cell.

Involved in the formation of primary lysosomes and enamel of tooth.

IV. Mitochondria

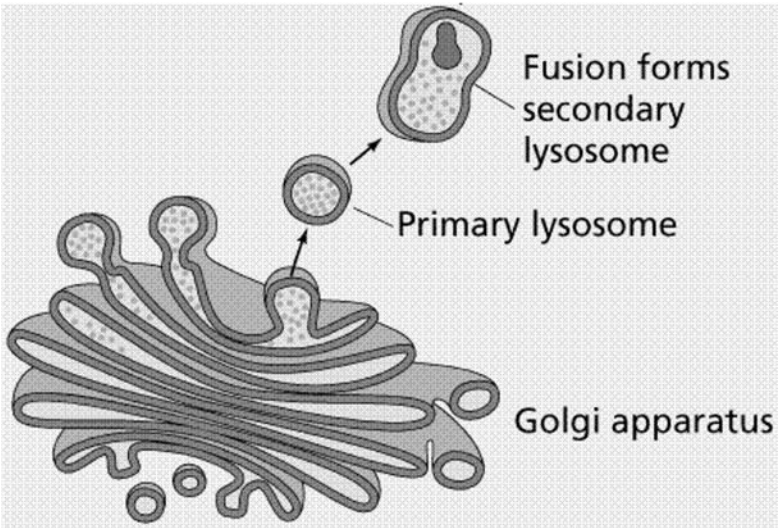


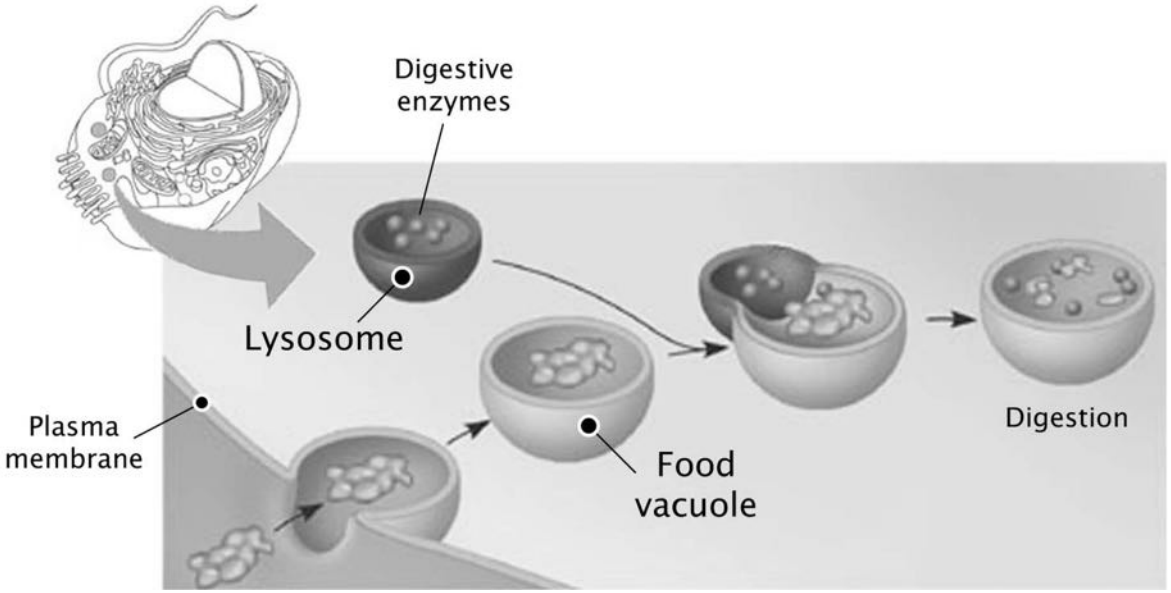
* Elongated or rounded contains their own DNA.
* Bounded by two membranes, its outer layer is smooth while inner membrane folds into cristae, where ATP generated.
* Ribosomes and mitochondrial DNA are found in the Matrix.

Functions of Mitochondria

* ATP generation.
* Oxidative phosphorylation.
* Catalyze the enzymatic reactions.

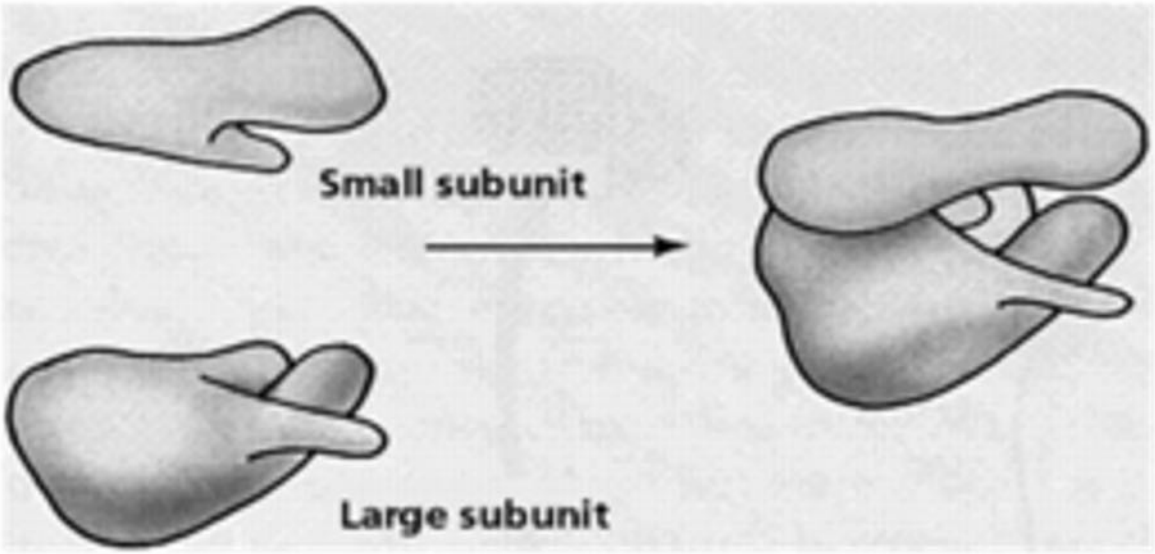
V. Lysosomes





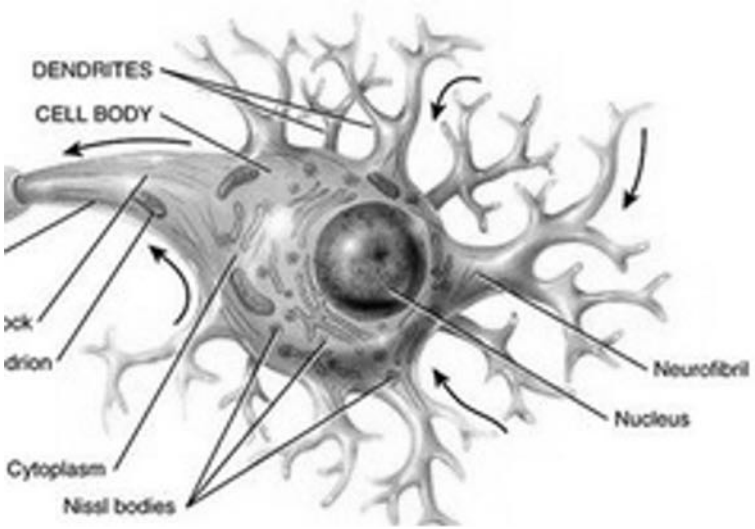
- Formed from interaction of ER and Golgi bodies and contains many hydrolytic enzymes.
- Digestion of certain materials or waste products (e.g. old mitochondria).
- Protecting cells from invading organisms.

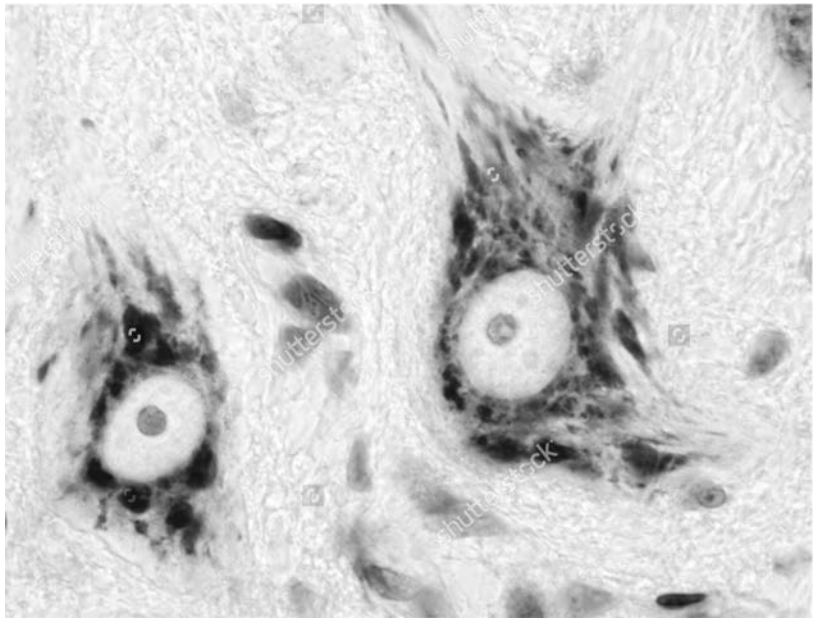
VI. Ribosomes



- Nucleoproteins (RNA + Proteins) formed by the nucleus , found in all cells except mature RBCs.
- Free or attached to ER.
- The smaller unit binding to mRNA, The larger unit has 2 binding sites for t-RNA.

Nissl bodies





- In cytoplasm and dendrites of nerve cells.
- Consisting of parallel arrays of cistena of r-ER.
- Treating nerve cell injury and react to prolonged stimulation of neurons.